

LOGICAL DEDUCTION]

V

1. LOGIC

The word 'Logic*' is derived from the Greek noun logos' meaning both 'thought*' and 'the word expressing thought'.

Thus, LOGIC is the 'science of thought as expressed in language'. This means that the questions on logic are to be solved as per the information given without any concern of the formal validity or truth of the statements i.e. conclusion should follow directly from the statements given.

With this unique characteristic, the Logic Test becomes an instrument of teaching the candidates to follow the rules and work as per the instructions without an error. Thus, it prepares the mind for all types of reasoning practices and teaches how to detect and avoid mistakes in the same.

LOGICAL REASONING

In Logic, any statement is termed as the **Proposition**. Thus, *a Proposition is a statement expressing certain relation between two or more terms*, analogous to a sentence in grammar.

The Proposition consists of three parts :

1. **Subject** : The Subject is that about which something is said.
2. **Predicate** : The Predicate is the part of the Proposition denoting that which is affirmed or denied about the subject.
3. **Copula** : The Copula is that part of the Proposition which denotes the relation between the Subject and the Predicate.

Consider the Proposition *%Man is cultured*

Here an information is given about the man. So 'Man*' is the Subject.

'Cultured'¹ is the quality affirmed for this Subject. So it is the Predicate.

'is' denotes the relation between the Subject and the Predicate. So, it is the Copula.

Four Fold Classification of Propositions :

'Propositions can be classified into four types ;

1. **Universal Affirmative Proposition** (denoted by A) : It distributes *only the subject i.e.*, the Predicate is not interchangeable with the subject while maintaining the validity of the Proposition, *e.g.*.

All men are animals.

This is Proposition A since we cannot say 'All animals are men.'

2. **Universal Negative Proposition** (denoted by E) : It distributes *both the Subject and the Predicate i.e.*, an entire class of predicate term is denied to the entire class of the subject term, as in the proposition, *e.g.*

No boy is intelligent.

3. **Particular Affirmative Proposition** (denoted by I) : It distributes *neither the Subject nor the Predicate, e.g.*

Some people are foolish.

Here, the subject term 'Some people'⁹ is used not for all but only for some men and similarly the predicate term 'foolish' is affirmed for a part of subject class. So, both are undistributed.

4. Particular Negative Proposition (denoted by O) :

It distributes *only the Predicate*, e.g.,

Some animals are not wild.

Here the subject term 'some animals' is used only for a part of its class and hence is undistributed while the predicate term 'wild' is denied in entirety to the subject term and hence is distributed.

These facts can be summarised as follows :

Proposition	Type
(a) (A) distributes subject only.	All S is F.
(b) (E) distributes subject and predicate both.	No S is P.
(c) (I) distributes neither.	Some S is P.
(d) (O) distributes predicate only.	Some S is not P.

SYLLOGISM : In Logic, we are required to deal with a particular type, termed as Syllogism. It was introduced by Aristotle.

In Syllogism, a *conclusion* has to be drawn from two propositions, referred to as the Premises.

Example : 1. All lotus are flowers.
2. All flowers are beautiful.
3. All lotus are beautiful.

Clearly, the propositions 1 and 2 are the Premises and the proposition 3, which follows from the first two propositions, is called the Conclusion.

Term : In Logic, a TERM is a word or a combination of words, which by itself can be used as a subject or predicate of a proposition.

Syllogism is concerned with three terms :

1. **Major Term** : It is the *predicate of the conclusion* and is denoted by P (first letter of 'Predicate').

2. **Minor Term** : It is the *subject of the conclusion* and is denoted by S (first letter of 'Subject').

3. **Middle Term** : It is the *term common to both the premises* and is denoted by M (first letter of 'Middle').

Note that the middle term does not occur in the conclusion.

Example : Premises : 1. All dogs are animals.
2. Tiger is a dog.

Conclusion : Tiger is an animal.

Here, 'animal' is the predicate of the conclusion and so, it is the Major Term, P. 'Tiger' is the subject of the conclusion and so, it is the Minor Term, S.

'Dog' is the term common to both the premises and so, it is the Middle Term, M.

Major and Minor Premise : Of the two premises, the *major premise* is that in which the middle term is the subject and the *minor premise* is that in which the middle term is the predicate.

Rules for deriving the conclusion :

1. The **conclusion does not contain the middle term.**

Example : Statements : 1. All men are girls.
2. Some girls are students.

- Conclusions :* 1. All girls are men.
2. Some students are girls.

Since both the conclusions 1 and 2 contain the middle term 'girls', so neither of them can follow,

2. *No term can be distributed in the conclusion unless it is distributed in the premises.*

Example : *Statements :* 1. Some dogs are goats.
2. All goats are cows.

- Conclusions :* 1. All cows are goats.
2. Some dogs are cows.

Statement 1 is an I type proposition which distributes neither the subject nor the predicate. Statement 2 is an A type proposition which distributes the subject. i.e. 'goats' only.

Conclusion 1 is an A type proposition which distributes the subject 'cow' only.

Since the term 'cow' is distributed in conclusion 1 without being distributed in the premises, so conclusion 1 cannot follow.

3. *The middle term (St) should be distributed at least once in the premises. Otherwise, the conclusion cannot follow.*

For the middle term to be distributed in a premise,

- (i) M must be the Subject if premise is an A proposition.
- (ii) M must be Subject or Predicate if premise is an E proposition.
- (iii) M must be Predicate if premise is an O proposition.

Note that in an I proposition, which distributes neither the Subject nor the Predicate, the middle term cannot be distributed.

Example : *Statements :* 1. All fans are watches.
2. Some watches are black.

- Conclusions :* 1. All watches are fans.
2. Some fans are black.

In the premises, the middle term is 'watches'. Clearly, it is not distributed in the first premise which is an A proposition as it does not form its subject. Also, it is not distributed in the second premise which is an I proposition. Since the middle term is not distributed at least once in the premises, so no conclusion follows.

4. *No conclusion follows*

(a) *if both the premises are particular*

Example : *Statements :* 1. Some books are pens.
2. Some pens are erasers.

- Conclusions :* 1. All books are erasers.
2. Some erasers are books.

Since both the premises are particular, no conclusion follows.

(b) *if both the premises are negative*

Example : *Statements :* 1. No flower is mango.
2. No mango is cherry.

- Conclusions :* 1. No flower is cherry.
2. Some cherries are mangoes.

Since both the premises are negative, neither conclusion follows.

(c) if the major premise is particular and the minor premise is negative.

Example : Statements : 1. Some dogs are bulls.

2. No tigers are dogs

Conclusions : 1. No dogs are tigers.

2. Some bulls are tigers.

Here the first premise containing the middle term 'dogs' as the Subject is the major premise and the second premise containing the middle term 'dogs' as the Predicate is the minor premise. Since the major premise is particular and the minor premise is negative, so no conclusion follows.

5. If the middle term is distributed twice, the conclusion cannot be universal.

Example : Statements : 1. All fans are chairs.

2. No tables are fans.

Conclusions : 1. No tables are chairs.

2. Some tables are chairs.

Here, the first premise is an A proposition and so, the middle term 'fans' forming the subject is distributed. The second premise is an E proposition and so, the middle term 'fans' forming the predicate is distributed. Since the middle term is distributed twice, so the conclusion cannot be universal.

6. If one premise is negative, the conclusion must be negative.

Example : Statements : 1. All grasses are trees.

2. No tree is shrub.

Conclusions : 1. No grasses are shrubs.

2. Some shrubs are grasses.

Since one premise is negative, the conclusion must be negative. So, conclusion 2 cannot follow.

7. If one premise is particular, the conclusion is particular.

Example : Statements : 1. Some boys are thieves.

2. All thieves are dacoits.

Conclusions : 1. Some boys are dacoits.

2. All dacoits are thieves.

Since one premise is particular, the conclusion must be particular. So, conclusion 2 cannot follow.

8. If both the premises are affirmative, the conclusion would be affirmative.

Example : Statements : 1. All women are mothers.

2. All mothers are sisters.

Conclusions : 1. All women are sisters.

2. Some women are not sisters.

9. If major premise be affirmative, the conclusion must be particular.

Example : Statements : 1. All plays are stories.

2. Some poems are plays.

Conclusions : 1. Some poems are stories.

2. All stories are poems.

The first premise containing the middle term 'plays' as the subject is the major premise. Also, it is affirmative. So, the conclusion must be particular. Hence, conclusion 2 cannot follow.

EXERCISE 1A

Directions : In each question below are given two statements followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusions and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Give answer (a) if only conclusion I follows; (b) if only conclusion II follows; (c) if either I or II follows; (d) if neither I nor II follows and (e) if both I and II follow.

1. **Statements** : All planets are moons.
All moons are stars. (Bank P.O. 1996)
Conclusions : I. All moons are planets.
II. All planets are stars.
2. **Statements** : All men are dogs.
All dogs are cats. (M.BA 1997)
Conclusions : I. All men are cats.
II. All cats are men.
3. **Statements** : All tubes are handles.
All cups are handles. (Bank P.O. 1997)
Conclusions : I. All cups are tubes.
II. Some handles are not cups.
4. **Statements** : All bags are cakes.
All lamps are cakes.
Conclusions : I. Some lamps are bags.
II. No lamp is bag.
5. **Statements** : All flowers are stems.
All stems are roots. (Bank P.O. 1993)
Conclusions : I. All roots are flowers.
II. All stems are flowers.
6. **Statements** : All puppets are dolls.
All dolls are toys.
Conclusions : I. Some toys are puppets.
II. All toys are puppets.
7. **Statements** : All apples are oranges.
Some oranges are papayas. (M.B JL 1998)
Conclusions : I. Some apples are papayas.
II. Some papayas are apples.
8. **Statements** : Some players are singers.
All singers are tall. (Bank P.O. 1998)
Conclusions : I. Some players are tall.
II. All players are tall.
9. **Statements** : All coins are crows.
Some crows are pens. (Bank P.O. 1997)
Conclusions : I. No pen is coin.
II. Some coins are pene.

10. **Statements** : All men are married.
Some men are educated. **(M.B.A. 1997)**
- Conclusions** ; I. Some married are educated.
II. Some educated are married.
11. **Statements** : All cars are tables.
Some children are tables. **(Hotel Management, 1992)**
- Conclusions** : I. Some cars are children.
II. Some children are cars.
12. **Statements** 2 All windows are needles.
Some trees are windows. **(Bank P.O. 1996)**
- Conclusions** : I. Some trees are needles.
II. Some trees are not needles.
13. **Statements** : Some dogs bark.
All dogs bite.
- Conclusions** : I. Those dogs who do not bark, also bite.
II. Those dogs who do not bark, not necessarily bite.
14. **Statements** : Some fools are intelligent.
Some intelligent are great, **(Bank P.O. 1998)**
- Conclusions** s I. Some fools are great.
II. All great are intelligent.
15. **Statements** : Some papers are files.
Some files are pens. **(Bank P.O. 1997)**
- Conclusions** : I. Some files are not pens.
II. Some pens are papers.
16. **Statements** : Some bottles are pencils.
Some pencils are glasses. **(Bank P.O. 1996)**
- Conclusions** : I. No glass is bottle.
II. Some bottles are glasses.
17. **Statements** : Some soldiers are famous.
Some soldiers are intelligent.
- Conclusions** : I. Some soldiers are either famous or intelligent.
II. Some soldiers are neither famous nor intelligent.
18. **Statements** : All boys are honest.
Sachin is honest.
- Conclusions** : I. Sachin is a boy.
II. All honest persons are boys.
19. **Statements** : Lawyers married only fair girls.
Shobha is very fair.
- Conclusions** : I. Shobha was married to a lawyer.
II. Shobha was not married to a lawyer.
20. **Statements** : Sohan is a good sportsman.
Sportsmen are healthy.
- Conclusions** : I. All healthy persons are sportsmen.
II. Sohan is healthy.

17. **Statements** All students in my class are intelligent.
Rohit is not intelligent. **(Asstt. Grade, 1992)**
- Conclusions** I. Rohit is not a student of my class.
II. Rohit must work hard.
22. **Statements** All hill stations have a sun-set point.
X is a hill station.
- Conclusions** I. X has a sun-set point.
II. Places other than hill stations do not have sun-set points.
23. **Statements** : Some sticks are bolts.
Kite is a stick.
- Conclusions** I. Some bolts are sticks.
II. Some kites are bolts.
24. **Statements** Some men are educated.
Educated persons prefer small families.
- Conclusions** I. All small families are educated.
II. Some men prefer small families.
25. **Statements** Some nurses are nuns.
Madhu is a nun. **(M.B.A. 1998)**
- Conclusions** I. Some nuns are nurses.
II. Some nurses are not nuns.
26. **Statements** All lamps are hooks.
No hook is coloured. **(Bank P.O. 1996)**
- Conclusions** I. Some lamps are coloured.
II. No lamp is coloured.
27. **Statements** All windows are doors.
No door is wall.
- Conclusions** I. No window is wall.
II. No wall is door.
28. **Statements** All locks are keys.
No key is a spoon.
- Conclusions** I. No lock is a spoon.
II. No spoon is a lock.
29. **Statements** All young scientists are open-minded.
No open-minded men are superstitious.
- Conclusions** I. No scientist is superstitious.
II. No young people are superstitious.
30. **Statements** All plants are trees.
No tree is green.
- Conclusions** I. Some plants are green.
II. Those plants which are not trees are green.
31. **Statements** No magazine is cap.
All caps are cameras. **(Bank P.O. 1997)**
- Conclusions** I. No camera is magazine.
II. Some caps are magazines.

32. **Statements** Some shirts are biscuits.
 No biscuit is book.
 Conclusions I. Some shirts are books.
 II. Some books are biscuits.
33. **Statements** Some books are pens.
 No pen is pencil. (Bank P.O. 1998)
 Conclusions I. Some books are pencils.
 II. No book is pencil.
34. **Statements** No women can vote.
 Some women are politicians.
 Conclusions I. Male politicians can vote.
 II. Some politicians can vote.
35. **Statements** Some books are toys.
 No toy is red. (Bank P.O. 1997)
 Conclusions I. Some books are red.
 II. Some books are not red.
36. **Statements** All birds are dogs.
 Some dogs are cats. (C.B.I. 1997)
 Conclusions I. Some cats are not dogs.
 II. All dogs are not birds.
37. **Statements** Many books are rocks.
 All rocks are clips.
 Conclusions I. Some books are clips.
 II. No rock is a book.
38. **Statements** Most clocks are fans.
 Some fans are walls.
 Conclusions I. Some walls are fans.
 II. Some clocks are walls.
39. **Statements** No man is a donkey.
 Rahul is a man.
 Conclusions I. Rahul is not a donkey.
 II. All men are not Rahul.
40. **Statements** : All poles are guns.
 Some boats are not poles. (M.B.A 1997)
 Conclusions I. All guns are boats.
 II. Some boats are not guns.

ANSWERS

1. (6): Since both the statements are affirmative, the conclusion must be affirmative. However, conclusion I cannot follow as it contains the middle term. So, only conclusion II follows.
2. (a): Since both the premises are affirmative, the conclusion must be affirmative. However, conclusion II being an A type proposition, distributes the term 'goats'. Since the term 'goats*' is distributed in II without being distributed in any of the premises, so conclusion II cannot follow. Thus, only I follows.

3. *id*): Both the premises are A type propositions. So, in either, the middle term 'handles' forming the predicate is not distributed.
Since the middle term is not distributed even once in the premises, so no conclusion follows.
4. (d) : Both the premises being A type propositions, the middle term 'cakes' forming the predicate is not distributed in any of them.
Since the middle term is not distributed even once in the premises, so no conclusion follows.
5. (d): Conclusion I being an A type proposition, distributes the term 'roots'. Since the term 'roots' is distributed in I without being distributed in any of the premises, so conclusion I cannot follow. Conclusion II cannot follow as it contains the middle term.
6. (a): Conclusion II, being an A type proposition, distributes the term 'toys'. Since the term 'toys' is distributed in II without being distributed in any of the premises, so conclusion II cannot follow. So, only I follows.
7. (d) : The first premise is A type and distributes the subject. So, the middle term 'oranges' which forms its predicate, is not distributed
The second premise is I type and does not distribute either subject or predicate. So, the middle term 'oranges' forming its subject is not distributed
Since the middle term is not distributed even once in the premises, so no conclusion follows.
8. (a) : Since one premise is particular, the conclusion must be particular. So, only conclusion I follows.
9. *Id*) : Since the middle term 'crows' is not distributed even once in the premises, so no conclusion follows.
10. (e) : Since one premise is particular, the conclusion must be particular. So, both I and II follow.
11. *id*): The first premise is an A type proposition. So, the middle term 'tables' forming the predicate is not distributed.
The second premise is an I type proposition. So, the middle term forming the predicate is not distributed. Since the middle term is not distributed even once in the premises, so no conclusion follows.
12. (a) : Since one premise is particular, so the conclusion must be particular. Also, since the term 'needles' is distributed in II (O type proposition) without being distributed in the premises, so, conclusion II cannot follow. Thus, only I follows.
13. (a): Clearly, conclusion I follows from the statements.
14. (d) : Since both the premises are particular, no conclusion follows as the middle term is not distributed even once in the premises.
15. (d): Since both the premises are particular, so no conclusion follows.
16. *id*): Since both the premises are particular, so no conclusion follows.
17. (d): Since both the premises are particular, so no conclusion follows.
18. (d): Both the premises are A type propositions. So, the middle term 'honest' forming the predicate in each is not distributed in either.
Since the middle term is not distributed even once, no conclusion follows.
19. (c): The data does not mention whether all fair girls were married to lawyers. So, either of the two conclusions may follow.
20. (6) : Conclusion I cannot follow as it contains the middle term. So, only conclusion II follows.
21. (a): Since one premise is negative, the conclusion must be negative. So, only conclusion I follows.
22. (a): Since both the premises are affirmative, the conclusion must be affirmative. So, only conclusion I follows.

23. *id*): The middle term 'sticks' forming the subject is not distributed in the first premise which is an I type proposition. The middle term forming the predicate is not distributed in second premise as it is an A type proposition and distributes subject only. Since middle term is not distributed even once, no conclusion follows.
24. *ib*): Since one premise is particular, the conclusion must be particular. So, only conclusion II follows.
25. *id*): Since the middle term is not distributed even once in the premises, so no conclusion follows.
28. *ib*): Since one premise is negative, the conclusion must be negative. So, only conclusion II follows.
27. *ia*): Since one premise is negative, the conclusion must be negative. Conclusion II cannot follow as it contains the middle term. So, only conclusion I follows.
28. (a): Since one premise is negative, the conclusion must be negative. So, I follows. But the reverse is not necessarily true. So, II does not follow.
29. *id*): The subject in both the conclusions is vague. The true conclusion is 'No young scientist is superstitious'. So, neither conclusion follows.
30. (d): Since one premise is negative, the conclusion must be negative. So, neither conclusion follows.
31. *id*): Since one premise is negative, the conclusion must be negative. So, II cannot follow. Also, the term 'camera*' is distributed in I without being distributed in any of the premises. So, conclusion I also does not follow.
32. *id*): Since one premise is negative, so conclusion must be negative. So, neither I nor II follows.
33. (tf): Since one premise is negative, the conclusion must be negative. So, conclusion I cannot follow.
Since one premise is particular, the conclusion must be particular. Also, the term 'books*' is distributed in II without being distributed in any of the premises. So, II also cannot follow.
34. *id*): Since one premise is negative, the conclusion must be negative. So, neither conclusion follows.
35. (6): Since one premise is negative, the conclusion must be negative. So, I cannot follow. Thus, only II follows
36. (rf): Since the middle term is not distributed even once in the premises, no conclusion follows.
37. (a): Since the first premise is particular, the conclusion must be particular. So, only conclusion I follows.
38. *id*): Since both the premises are particular, no conclusion follows.
39. *ia*): Since one premise is negative, the conclusion must be negative. Conclusion II cannot follow as it contains the middle term. So, only conclusion I follows.
40. *id*): Clearly, the term 'guns*' is distributed in both the conclusions without being distributed in any of the premises. So, no conclusion follows.

EXERCISE 1B

Directions : In each questions below are given two statements followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusions and then decide which of the the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Give answer (a) if only conclusion I follows; (6) if only conclusion II follows; (c) if either I or II follows; (d) if neither I nor II follows and (e) if both I and II follow.

- Statements : All dogs are jackals.
Some jackals are crows.

Conclusions : I. Some dogs are crows.
II. All dogs are crows.
- (Bank P.O. 1994)
2. Statements : Some children are adults.
Some adults are old.

Conclusions : I. Some children are not old.
II. Some adults are not old.
3. Statements : AH keys are locks.
All locks are screws.

Conclusions : I. All screws are keys.
II. Some locks are keys.
- (Bank P.O. 1998)
4. Statements : All poets are readers.
No reader is wise.

Conclusions : I. No poet is wise.
II. All readers are poets.
5. Statements : Some kites are horses.
All horses are dogs.

Conclusions : I. All dogs are horses.
II. Some dogs are horses.
6. Statements : Some calendars are sticks.
No stick is flower.

Conclusions : I. Some calendars are flowers.
II. No calendar is flower.
- (Bank P.O. 1996)
7. Statements : Most crops are machines.
Some machines are fools.

Conclusions : I. Some fools are machines.
II. Some crops are fools.
8. Statements : Some flies are ants.
All insects are ants.

Conclusions : I. All flies are ants.
II. Some ants are insects.
9. Statements : Some stones are cups.
Some cups are black.

Conclusions : I. Some black are not cups.
II. Some cups are stones.
10. Statements : All goats are wolves.
Some wolves are tigers.

Conclusions : I. Some goats are tigers.
II. Tigers which are wolves are
- goats.
- (Bank P.O. 1992)
11. Statements : Some phones are watches.
All watches are guns.

Conclusions : I. All guns are watches.
II. Some guns are phones.

12. **Statements** All teachers are good.
Some women are teachers.
- Conclusions** I. All good teachers are women.
II. Some women are good.
13. **Statements** All roads are poles.
No pole is house.
- Conclusions** I. Some roads are houses.
II. Some houses are poles.
14. **Statements** Some pastries are toffees.
All toffees are chocolates.
- Conclusions** I. Some chocolates are toffees.
II. Some toffees are not pastries.
15. **Statements** Some chairs are stools.
Table is a chair.
- Conclusions** I. Some stools are chairs.
II. Table is not a stool.
16. **Statements** All tigers are ships.
Some ships are cupboards.
- Conclusions** I. Some tigers are cupboards.
II. Some cupboards are tigers.
17. **Statements** Some vegetables are fruits.
No fruit is black.
- Conclusions** I. Some fruits are vegetables.
II. No fruit is black.
18. **Statements** Some aeroplanes are living beings.
Some non-living beings are ghosts.
- Conclusions** I. Some aeroplanes are ghosts.
II. Some aeroplanes are not ghosts.
19. **Statements** All dresses are shoes.
No shoe is brown.
- Conclusions** : I. No dresses are brown.
II. Some shoes are dresses.
20. **Statements** Some boys are men.
No man is black.
- Conclusions** I. Some boys are not black.
II. Some men are boys.
21. **Statements** All stones are diamonds.
Some diamonds are pearls.
- Conclusions** I. Some pearls are stones.
II. All diamonds are pearls.
22. **Statements** Some parrots are crows.
No crow is green.
- Conclusions** I. No parrot is green.
II. No crow is white.

(R.B.I. 1997)

(Bank P.O. 1998)

(Bank P.O. 1997)

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5. *id*) : None of the conclusions follows as both contain the middle term.
6. *id*) : Since one premise is particular, the conclusion must be particular. So, II cannot follow.
Since one premise is negative, the conclusion must be negative. So, I cannot follow.
7. *id*) : Since both the premises are particular, no conclusion follows.
8. *id*) : Since the middle term is not distributed even once in the premises, no conclusion follows.
9. *id*) : Since both the premises are particular, no conclusion follows.
10. (ci) : Since the middle term is not distributed even once in the premises, no conclusion follows.
11. (6) : Since one premise is particular, the conclusion must be particular. So, only conclusion II follows.
12. (6) : Since one premise is particular, the conclusion must be particular. So, only conclusion II follows.
13. (d) : Since one premise is negative, the conclusion must be negative. So, neither conclusion follows.
14. Id) : Neither conclusion I nor II follows as both contain the middle term.
15. (d) : The first premise is an I type proposition. So, the middle term 'chairs' forming the subject is not distributed.
The second premise is an A type proposition. So, the middle term 'chairs*' forming the predicate is not distributed.
Since the middle term is not distributed even once, no conclusion follows.
16. *id*) : Since the middle term is not distributed even once in the premises, no conclusion follows.
17. *id*) : Since one premise is particular, the conclusion must be particular. So, II cannot follow.
Since one premise is negative, the conclusion must be negative. So, I cannot follow.
18. *id*) : Since both the premises are particular, no conclusion follows.
19. *ia*) : Since one premise is negative, the conclusion must be negative. So, only conclusion I follows.
20. *ia*) : Since one premise is negative, the conclusion must be negative. So, II does not follow and only I follows.
21. (d) : Since the middle term is not distributed even once in the premises, no conclusion follows.
22. *id*) : Since one premise is particular, the conclusion must be particular. So, neither I nor II follows.
23. (d) : Since the middle term is not distributed even once in the premises, no conclusion follows.
24. *id*) 25. (6) 26. (6) 27. (a) 28. *ia*) 29. *id*) 30. *ic*)

TYPE 2

(When more than two conclusions are given)

In this type of questions, two statements called premises are given, followed by four conclusions. The candidate is required to find out which of the conclusions logically follow from the given premises. More than one conclusion may also follow. In such questions, first the given statements are analysed. If the middle term is not distributed even once, no conclusion follows. Thus, the middle term must be distributed at least once. To derive the correct conclusions, we usually take the help of Venn diagrams. Also, no conclusion follows in the following cases :

- (i) If both the premises are particular
- (ii) If both the premises are negative

However, in some cases, more than one Venn diagrams may be possible. In such cases, all the possible Venn diagrams are drawn and the solution is derived from each of these separately. Finally, the solution common to all the diagrams is taken as the answer.

ILLUSTRATIVE EXAMPLES

Ex. I. Statements : All books are cakes.
All cakes are apples.

Conclusions : I. Some cakes are books.
II. No cake is book.
III. Some apples are books.
IV. All apples are books,

- (a) Only I follows (6) Only either I or II follows
(c) Only I and III follow (rf) Only either III or IV follows
(e) None follows

Sol. For the given statements, the Vern diagram is as shown where B indicates books, C indicates cakes and A indicates apples.

Now, since B has a common area with C, so I follows while II does not. Also, B has a common area with A but does not entirely lie within A. So III follows while IV does not.

Thus, only I and III follow.

Hence, the answer is (c).

Ex. 2. Statements ; Some flowers are apples.
Some apples are stones.

Conclusions : I. No flower is stone.
II. All apples are stones.
III. Some stones are flowers.
IV. No apple is flower,

- (a) Only either I or III follows (6) Only I and IV follow
(c) Only II and III follow (id) Only I, III and rV follow
(e) None follows

Sol. Since both the statements or premises are particular, no conclusion follows.

Hence, the answer is (e).

Ex. 3. Statements s All leaves are papers.

Some papers are pictures.

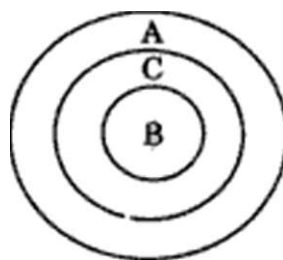
Conclusions : I. All pictures are papers.
II. All pictures are leaves.
III. Some leaves are pictures.
IV. No leaf is a picture.

- (a) Only I, III and IV follow (6) Only II and III follow
(c) Only III and IV follow (id) Only III follows
(e) None of these

Sol. Clearly, here the middle term is 'papers'. Now, the first premise is an A type proposition and distributes the subject only. So, the middle term 'papers' forming the predicate, is not distributed. The second premise is an I type proposition and distributes neither the subject nor the predicate. So, the middle term 'papers' forming the subject, is not distributed.

Since the middle term is not distributed even once in the premises, no conclusion follows.

Hence, the answer is (e).



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Statements : All windows are rods.
Some rods are frames.

Conclusions : I. **All** frames are rods.
 ' II. All frames are windows.
 III. Some windows are frames.
 IV. No window is a frame.

ia) Only I follows ib) Only II and III follow
 ic) Only either II or III follows id) Only either I or IV follows
 ie) None follows

Statements : Some clothes are marbles.
Some marbles are bags.

Conclusions : I. No cloth is a bag.
II. All marbles are bags.
III. Some bags are clothes.
IV. No marble is a cloth.

(a) Only either I or IV follows
 ic) Only either I or III follows
 ie) All follow

(6) Only either I or II follows
 id) None follows

Statements : Some pillows are curtains.
No curtain is a table.

Conclusions :

- I. No pillow is a table.
- II. Some pillows are not tables.
- III. Some curtains are pillows.
- IV. No curtain is a pillow.

ia) Only I and III follow
 ic) None follows
 (e) Only either I or IV follows

Statements : Some frogs are bricks.
All bricks are cakes.

Conclusions :

- I. Some cakes are not frogs.
- II. Some cakes are frogs.
- III. No cake is frog.
- IV. All frogs are cakes.

ia) None follows
 ic) Only I, II and IV follow
 ie) All follow

Statements : No parrot is crow.
All crows are bats.

Conclusions : I. Some bats are parrots.
II. All bats are parrots.
III. Some bats are crows.
IV. Some bats are not crows.

ia) None follows
 ic) Only I, II and III follow
 ie) Only III and IV follow

(S.B.I.P.O.1997)

(Bank P.O. 1994)

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- 12. Statements** : No educationists are researchers.
All researchers are teachers. (Bank P.O. 1996)
- Conclusions** : I. No teacher **is** researcher.
II. No teacher is educationist.
III. Some researchers are teachers.
IV. Some teachers are researchers.
- (a) Only II follows (6) None follows
(c) Either I or III follows (d) Only III and IV follow
(e) All follow
- 13. Statements** : All bags are chalks.
All chalks are bottles. (S.B.I.P.O. 1997)
- Conclusions** : I. Some bottles are bags.
II. All bags are bottles.
III. All bottles are bags.
IV. Some chalks are not bags.
- (a) Only I, II and IV follow (6) Only I III and IV follow
(c) Only II, III and IV follow (d) All follow
(e) None of these
- 14. Statements** : No fan is shirt.
All shirts are trousers.
- Conclusions** : I. All fans are trousers.
II. No fan is trouser.
III. Some trousers are shirts.
IV. All trousers are shirts,
- (a) Only I follows (6) Only II follows
(c) Only either I or II follows (d) Only III and IV follow
(e) None of these
- 15. Statements** : Some cubs are tigers.
Some tigers are goats. (Bank P.O. 1994)
- Conclusions** : I. Some cubs are goats.
II. No cub is goat.
III. All cubs are goats.
IV. All goats are cubs,
- (a) Only either I or II follows (6) Only either II or III follows
(c) Only either I or IV follows (d) Only either II or IV follows
(e) None of these
- 16. Statements** : Some coolers are watches.
No watch is bed.
- Conclusions** : I. No watch is cooler.
II. No cooler is watch.
III. Some watches are beds.
IV. Some coolers are beds.
- (a) None follows (6) Only I and IV follow
(c) Only either II or III follows (d) Only either III or IV follows
(e) Only either II or IV follows

- Directions** (Questions 20 to 26) : In each question below are given two statements followed by five conclusions numbered (a), (b), (c), (d) and (e). You have to take the two given statements to be true even if they seem to be at variance from the commonly known facts. Read the **conclusions** and then decide which of the given conclusions logically follows from the two given statements.

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- 28. Statements** All teachers are doctors.
All doctors are engineers.
All engineers are students. **(Bank P.O. 1995)**
- Conclusions** I. Some students are teachers.
II. All doctors are students.
III. Some engineers are teachers.
IV. All doctors are teachers,
- (a) Only I and II follow (6) Only I and III follow
(c) Either I or II, and III follow (d) Either II or IV follows
(e) None of these
- 29. Statements** : Some birds are insects.
All birds are butterflies.
All insects are snakes. **(S.B.I.P.O. 1995)**
- Conclusions** : I. Some snakes are birds.
II. Some butterflies are insects.
III. Some snakes are butterflies.
IV. Some insects are birds.
- (a) None follows (6) Either I or III follows
(c) All follow (d) Only IV follows
(e) None of these
- 30. Statements** Some bananas are apples.
All apples are tomatoes.
Some potatoes are tomatoes.
- Conclusions** I. Some bananas are tomatoes.
II. Some potatoes are bananas.
III. Some apples are potatoes.
IV. Some apples are bananas.
- (a) Only I follows (6) Only I and IV follow
(c) Only I and IV follow (d) Either II or III, and I follow
(e) None of these
- 31. Statements** All boxes are pans.
Some boxes are jugs.
Some jugs are glasses.
- Conclusions** : I. Some glasses are boxes.
II. No glass is box.
III. Some jugs are pans.
IV. No jug is pan.
- (a) Only I and II follow (6) Either I or II, and III follow
(c) Only III follows (d) Either I or II, and either III or IV follow
(e) None of these
- 32. Statements** : Some books are papers.
Some papers are cars.
No car is white. **(Bank P.O. 1996)**

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- 2. (e): Here the first premise is an A type proposition and distributes only the subject So, the middle term 'rods' forming its predicate is not distributed. The second premise is an I type proposition and distributes neither the subject nor the predicate So, the middle term 'rod' forming the subject is not distributed. Since the middle term is not distributed even once in the premises, no conclusion follows.
- 3. (d): Since both the premises are particular, no conclusion follows.
- 4. *ib*): Clearly, for the given data two Venn diagrams A and B are possible.

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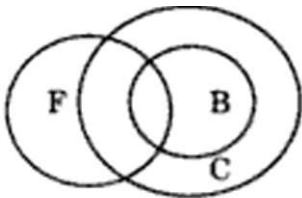
A

G 8 S

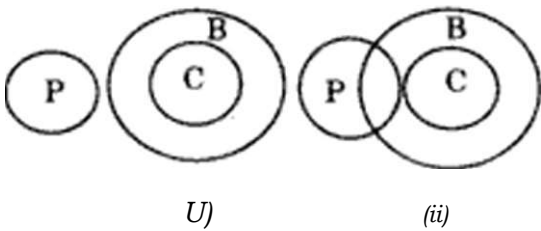
B

From diagram A, conclusions I, II and III follow.
From diagram B, conclusions II and III follow.
The solution common to the two diagrams is : II and III follow.

- 5. (6): Clearly, it follows from the Venn diagram that some cakes are frogs and some are not *i.e.* conclusions I and II follow but conclusions III and IV cannot follow.



- 6. *ie*): Clearly, two Venn diagrams (i) and *iii*) are possible as shown.
From diagram (i), conclusions III and IV follow.
From diagram (ii), conclusions I, III and IV follow.
The common solution is : Only III and IV follow.



- 7. (a): Clearly the area not common to Students and Brilliant will represent the dull students in the Venn diagram. So, conclusion I follows.
- 8. *ie*) Clearly, in the adjoining Venn diagram, since R and W are disjoint, so I and II follow while III does not.
Since C does not lie entirely within R, so IV does not follow.
- 9. (c) Clearly, two Venn diagrams (i) and (ii) are possible as shown

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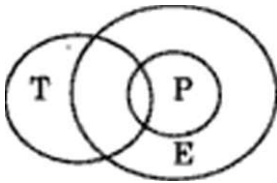
(i)

Q 8 S

(ii)

From (i), conclusions I and III follow.
From (ii), conclusions I, II and III follow.
The common solution is : Only I and III follow.

- 10. (c) Since E and T do not lie entirely within P, so neither I nor II follows. Since some area common to T and E lies outside P, so III follows.
Also, E and T have a common area.
So, IV also follows



11. (e): Since the middle term is not distributed even

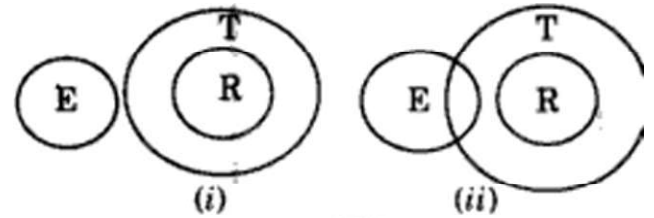
in the premises, no conclusion follow

12. (rf): Clearly, two Venn diagrams (i) and (ii) are possible as shown

From (i), II, III and IV follow.

From (ii), III and IV follow.

The common solution is : Only III and IV follow.



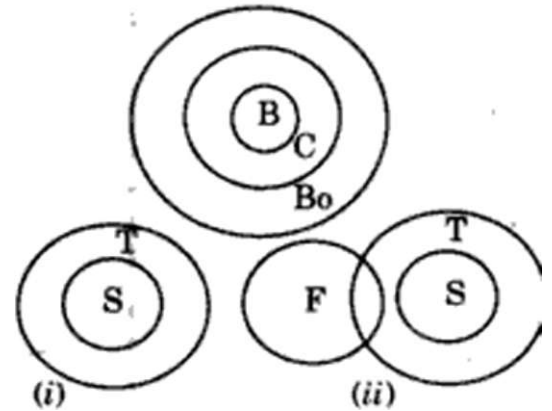
13. (a) Clearly, it follows from the Venn diagram that only conclusions I, II and IV follow.

14. (e) Clearly, two Venn diagrams (i) and (ii) are possible as shown.

From (i), II and III follow.

From (ii), only III follows.

The common solution is : Only III follows



15. (e): Since the two premises are particular, no conclusion follows.

16. (a): Clearly, two Venn diagrams (i) and (ii) are possible as shown

From (i), none of the conclusions follows.

From (ii), only IV follows.

Thus, the common solution is : None follows.

Q ® © (385
a) (ii)

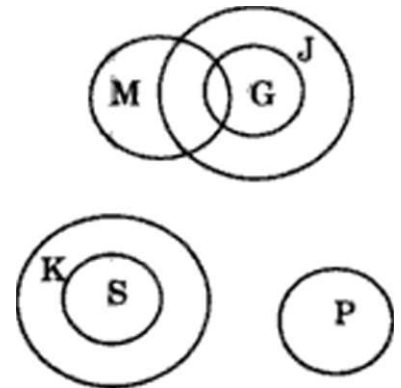
17. (e): Since J and M have a common area, so both I and II follow.

Since G and M have a common area, so IV follows.

However, since J does not lie entirely within G, so III does not follow.

18. (a): Since S and P are disjoint, so I follows while II and IV do not.

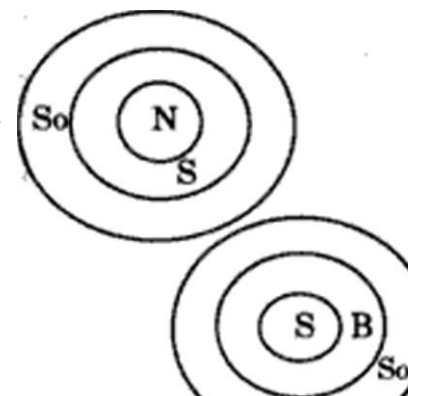
Since K does not lie entirely within S, so III does not follow.



18. (e): Since the middle term is not distributed even once in the premises, no conclusion follow

20. (a): Clearly, since the area for novels (N) lies entirely within the area for songs (So), it follows that all novels are songs. So, some songs are novels. But only one conclusion has to follow and the most logical conclusion is 'All novels are songs'. So, (a) follows.

21. (a) : Here S denotes shirts, B denotes belts and So denotes socks. Then, clearly conclusion (a) follows with the same explanation as in Q. 20.



22. (e): Since both the premises are particular, no conclusion follows.

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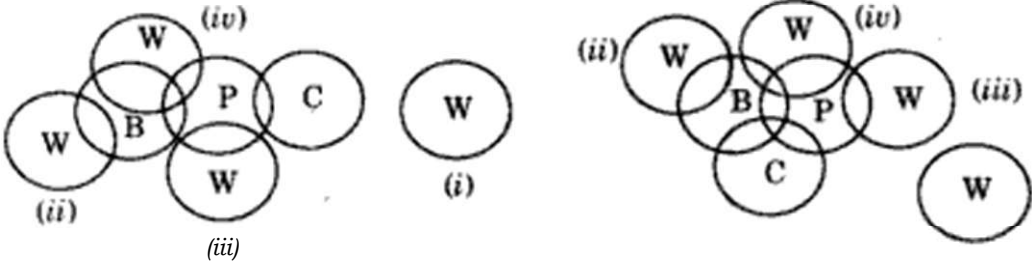
From (iii), conclusions I, II, III and IV follow.
The common solution is : Only I and IV follow.

31. (b) : Clearly, the following two Venn diagrams are possible :



From, (i), conclusions II and III follow
From (ii), conclusions I and III follow.
Combining the two, we have : Either I or II, and III follow.

32. (a) : Clearly two cases X and Y arise with possibilities (i), (ii), (iii) and (iv).



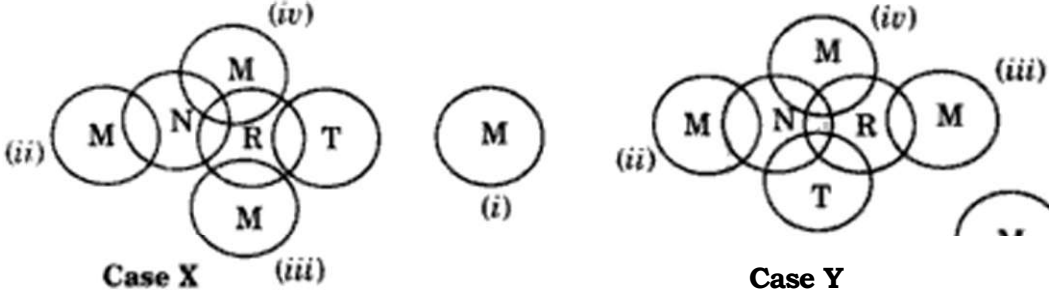
Case X
In both cases, we have :
From (i), only I and II follow
From (ii), only I, II and IV follow.
From (iii), only I and III follow.
From (iv), only I, III and IV follow.
The common solution in all the above is : Only I follows.

33. (d) : Clearly, the following Venn diagrams are possible :



From (i), conclusions II and III follow.
From (ii), conclusions I and III follow.
Combining the two, we get : Either I or II, and III follow.

34. (d) : Clearly, two cases X and Y arise with possibilities (i), (ii), (iii) and (iv).



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